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CHE 313

1/28/26

1.D1 (2)

1200 kmol/h

30.0 mol % ethanol

70.0 mol % water

$$W_E = 46.07 \frac{\text{kg}}{\text{kmol}}$$

$$W_W = 18.02 \frac{\text{kg}}{\text{kmol}}$$

$$F_{ZE} = 1200 \cdot 0.3 = 360 \frac{\text{kmol}}{\text{h}}$$

$$360 \cdot 46.07 = 16585 \frac{\text{kg}}{\text{h}}$$

$$F_{ZW} = 0.7 \cdot 1200 = 840 \frac{\text{kmol}}{\text{h}}$$

$$840 \cdot 18.02 = 15137 \frac{\text{kg}}{\text{h}}$$

$$F_{W_T} = F_{WE} + F_{WW} = 16585 + 15137 = 31722 \frac{\text{kg}}{\text{h}}$$

$$\frac{W_E}{W_{TOT}} = 0.523 = 52.3\%$$

$$F_{\text{total}} \cdot 2.20 = 69788.4 \frac{\text{kg}}{\text{h}}$$

1.D3

$$F_E = 31722 \frac{\text{kg}}{\text{h}}$$

$$\frac{360 \frac{\text{kmol}}{\text{h}}}{1200 \frac{\text{kmol}}{\text{h}}} = 0.3$$

$$\left. \begin{array}{l} 30.0 \text{ wt\%} / 46.07 = 0.006 \\ 70.0 \text{ wt\%} / 18.02 = 0.039 \end{array} \right\} 0.045 \frac{\text{kmol}}{\text{kg}} \rightarrow 22.22 \frac{\text{kg}}{\text{kmol}}$$

$$F = 1200 \cdot 22.22 = 26664 \frac{\text{kg}}{\text{h}}$$

$$F_E = 0.3 \cdot 26664 = 7999.2 \frac{\text{kg}}{\text{h}}$$

$$\frac{7999.2}{46.07} = 173.63 \frac{\text{kmol}}{\text{h}} \quad 77.$$

$$x_E = \frac{173.63}{1200} = 0.145$$

3. (2)

60.0 mol % methanol
40.0 mol % vaporized
 $F = 100 \frac{\text{kmol}}{\text{h}}$

$$V = 0.4 \cdot 100 = 40 \frac{\text{kmol}}{\text{h}}$$

$$L = 0.6 \cdot 100 = 60 \frac{\text{kmol}}{\text{h}}$$

$$Fz = Vy + Lx$$

$$\frac{V}{F} = \frac{40}{100} = 0.4$$

$$100 \cdot 0.6 = 40y + 60x$$

$$60 - 60x = 40y$$

$$y = 1.5 - 1.5x$$

$$y = kx$$

b.

$$V = 0.4 (1500) = 600 \frac{\text{kmol}}{\text{h}}$$

$$L = 0.6 (1500) = 900 \frac{\text{kmol}}{\text{h}}$$

$$1500 \cdot 0.6 = 600y + 900x \quad k_1 = \frac{y_1}{x_1}$$

$$900 - 900x = 600y$$

$$1.5 - 1.5x = y$$

$$y_1 = \frac{k_1 x_1}{1 + (k_1 - 1) \frac{V}{F}}$$

$$x_1 = \frac{z_1}{1 + (k_1 - 1) \frac{V}{F}}$$

c.

30.0 mol % methanol
20.0 mol % methanol

$$F = 1000 \frac{\text{kmol}}{\text{h}}$$

$$\frac{V}{F} ?$$

$$Fz$$

$$Fz = Vy + Lx$$

$$z = \frac{V}{F} y + \frac{L}{F} x$$

$$\frac{L}{F} = 1 - \frac{V}{F}$$

$$z = \frac{V}{F} y + (1 - \frac{V}{F}) x$$

$$z = 0.3$$

$$x = 0.2$$

$$y_1 = k_1 x_1$$

$$y = \frac{k_1 z_1}{1 + (k_1 - 1) \frac{V}{F}}$$

d. $x = 45 \text{ mol } \%$
 $L = 1500.0 \frac{\text{kmol}}{\text{h}}$
 $\frac{V}{F} = 0.2$

* don't know how to find k value

$$y = \frac{kx_i}{1 + (k_i - 1)x_i}$$

$$Fz = Vy + Lx$$

$$R = V + L$$

$$z = \frac{V}{F}y + \left(1 - \frac{V}{F}\right)x$$

= ??

4. ①

a. $y_M = z_M = 0.25$ $P = 1 \text{ atm}$
 $25 \text{ mol } \%$ $25 \text{ mol } \%$ methanol

b. $z_M = x_M = 25 \text{ mol } \%$

c.

$$0.25 = \frac{1}{3}y + \frac{2}{3}x$$

~~assuming~~ $R = 1$

$$z = \frac{V}{F}y + \left(1 - \frac{V}{F}\right)x$$

d. $0.25 = \frac{2}{3}y + \frac{1}{3}x$

